

Apache Airflow Contribution & Verification Agent Skills

Proof of Concept

- **Workflow prototype:** skill generation from contributing docs via pre-commit and generated `SKILL.md` artifacts:
 - <https://github.com/apache/airflow/compare/main...RoyLee1224:airflow:feat/contribution-skills>
- **Evaluation report:** benchmark design and eval harness summary
 - <https://github.com/RoyLee1224/airflow/blob/9f72eebd8a858a3afdc551092dd8a05d3328bd2/airflow-contribution-workspace/iteration-1/report.md>

About Me

I am a first-year NYU MSIS student with 56 merged PRs in Apache Airflow (all-time rank #81), including two PRs recognised as PR of the Month:

- [#50626](#) — Initiated i18n for Airflow UI — resolved a core issue open since 2020, enabling 400+ community PRs to expand language support to 17+ languages
- [#54252](#) — Add Calendar view for DAG — a major feature for the 3.1.0 release, built from scratch using React and TypeScript

My contributions span frontend, backend, dev tools, and providers, giving me a full-stack understanding of the codebase and contributor workflow. Through my own development sessions with AI agents, I have repeatedly observed the workflow gap this proposal targets. This project is a natural next step in deepening my understanding of Airflow internals and contributing sustainably to the community.

Background

What exists already: Airflow has `AGENTS.md`, which documents correct workflows for AI agents. Preliminary results in `report.md` show that single-prompt workflow selection is already fairly strong: in the initial routing evals, the baseline setup handled Helm and Postgres cases correctly, while the main remaining failure was incorrectly choosing `breeze` over `uv` for a `db_test`-only case.

The identified need: The real failure mode is **long-context degradation**. Chroma's 2025 study of 18 frontier models found that every model's output quality degrades as input length increases (Hong et al., 2025). In practical Airflow development sessions — involving multi-step debugging, cross-package changes, and error log analysis — agents accumulate context that pushes them into what practitioners call the "Dumb Zone" (Horthy, 2025): a state where the model reverts to generic Python project patterns rather than Airflow-specific workflows. This proposal treats that failure mode as a harness-design problem rather than only a prompt-writing problem. The goal is to build versioned skill artifacts, selective context loading, and reproducible evaluations so Airflow-specific workflows remain reliable over long-running sessions.

What can be reused:

- Existing `AGENTS.md` judgment rules and architecture boundaries
- Existing `contributing-docs/*.rst` as the single source of truth for commands

- Existing `prek` hook infrastructure for static checks

What needs to be created:

- A skills/ directory containing domain-specific `SKILL.md` files (e.g. testing, PR workflow, UI), auto-synced from contributing-docs
- A community agent-failure case collection mechanism
- A benchmark framework for evaluating agent workflow compliance

Design / Description of Work

Phase 1 — `SKILL.md` with Auto-Sync

Separate concerns currently mixed in `AGENTS.md`:

```
SKILL.template.md      ← human-maintained: workflow logic, env detection

contributing-docs/*.rst ← single source of truth: concrete commands

↓ prek hook

SKILL.md               ← generated output consumed by agents
```

- Add lightweight agent-skill-start/end markers to contributing-docs
- Implement `prek` hook to extract marked blocks and inject into template
- CI fails on drift, ensuring `SKILL.md` stays in sync
- Simplify `AGENTS.md` — move command tables to `SKILL.md`, keep judgment rules

Phase 2 — Community Failure Case Collection

This idea is partly inspired by how `protm` highlights especially valuable monthly contributor work. In a similar spirit, well-diagnosed agent failure cases can become a maintainable source of project knowledge: they surface recurring workflow mistakes, clarify where agent guidance is incomplete, and provide concrete inputs for improving `AGENTS.md`, `SKILL.md`, and future benchmark scenarios over time.

I will propose a `kind:agent-case-study` PR label, consistent with existing conventions, together with an optional PR template section for contributors who used AI tools. If the approach is accepted by the community, a lightweight GitHub Action can aggregate tagged reports into `dev/agent-cases/YYYY-MM.md`.

```
## AI Agent Experience (optional)
```

- Tool used:
- What went wrong:
- Command agent used incorrectly:
- Correct command:
- Approximate session length:

Phase 3 — Airflow-Specific Benchmark

Using real merged PRs as ground truth, I will construct reproducible benchmark scenarios by reverting selected changes to a pre-PR state and asking an agent to complete the task from the original problem description.

Each benchmark run will capture:

- the prompt and task description
- the commands chosen by the agent
- the execution trace
- the resulting artifacts

Compliance will be scored through a combination of deterministic checks and rubric-based evaluation. Initial scoring checkpoints include correct `uv` vs `breeze` routing, `--project` correctness, backend selection, fallback behavior on environment failure, and commit-message format.

Note: For the POC, I am using [Claude's eval workflow](#) as a convenient prototype harness. The purpose is not to depend on a Claude-specific evaluation system long-term, but to validate the approach before turning them into a project-specific framework for Apache Airflow.

Phase 4 — Modular Skills Folder

```
.agents/skills/  
  
    airflow-contribution/  
  
    airflow-ui/  
  
    airflow-test/
```

Split `SKILL.md` into focused skills that load only what is relevant, reducing context pressure.

Results for the Apache Community

- AI agents become reliably Breeze-aware across realistic multi-turn development sessions, not just single prompts
- Contributing-docs becomes the enforced single source of truth for both human contributors and AI agents
- Community-sourced failure cases create a feedback loop for continuously improving `AGENTS.md` and `SKILL.md`
- The benchmark provides a reproducible way to evaluate any future changes to agent guidance, and can be adopted by other Apache projects facing the same problem

Deliverables

Milestone	Deliverable
Week 2	Initial <code>SKILL.template.md</code> , generated <code>SKILL.md</code> , CI drift detection, and tests
Week 4	Refined skill-generation pipeline and proposal draft for community feedback on agent case collection
Week 8	Benchmark harness with 3 initial routing scenarios and scoring outputs

Milestone	Deliverable
Week 10	Seed set of failure cases and benchmark refinements based on observed failure patterns
Week 12	Documentation and modular skills structure for at least 3 task-specific skill domains

Other Commitments

Semester ends May 5th, three weeks before coding period begins May 26th. No academic obligations during GSoC coding period.

Community Engagement

- Discussed proposal direction with mentor Jason before submission
- Active on Airflow Slack — will use it for day-to-day questions and quick feedback alongside the dev mailing list
- `kind:agent-case-study` label and PR template changes require community consensus — plan to propose on mailing list in Week 1
- Code reviews will be requested on each phase before moving to the next
- Progress updates posted to dev mailing list at each milestone
- All work submitted as PRs to apache/airflow, following existing contribution guidelines